



Society of Petroleum Engineers

SPE-227402-MS

Innovative Integration of Cutting Analysis and LWD Data for Geosteering Optimization

S. Rashidi Aghdam and D. Ignatov, ADAGA Solutions LTD, Muscat, Oman; V. Voynilenko and A. G. Manikin, ADAGA Solutions LTD, Belgrad, Serbia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227402-MS](https://doi.org/10.2118/227402-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Objectives/Scope: This paper presents a groundbreaking approach to geosteering optimization by integrating real-time analysis of drilling cuttings with Logging While Drilling (LWD) data. The primary objective is to enhance the precision of wellbore placement within geologically complex reservoirs, a critical factor in modern drilling operations. By improving the accuracy of well trajectories, this technique aims to maximize reservoir contact, increase hydrocarbon recovery, reduce operational risks, and enhance overall drilling efficiency. Geosteering, the process of dynamically adjusting a well's path based on real-time subsurface data, has become indispensable with the rise of horizontal and extended-reach wells, where even minor deviations can significantly impact production outcomes.

The methodology combines geochemical data from portable X-ray fluorescence (XRF) spectrometry with conventional LWD measurements, such as gamma ray, resistivity, density, and neutron porosity logs. This fusion provides a multidimensional, high-resolution view of the subsurface, enabling drilling teams to detect subtle lithological changes and make informed steering decisions. The scope of this study encompasses both the development of this integrated approach and its practical validation through field trials conducted in the "Uslovnoye" oil field in Western Siberia. Known for its challenging terrigenous reservoirs with rapid lithological variations and structural complexities, this field serves as an ideal testbed for evaluating the technique's effectiveness in real-world conditions.

This approach addresses longstanding challenges in geosteering, particularly in environments where traditional methods falter due to limited resolution or interpretive ambiguity. By offering a real-time, data-driven solution, it promises to transform how operators navigate complex geological formations, delivering measurable improvements in operational and economic performance.

Methods, Procedures, Process: The methodology hinges on a streamlined workflow that integrates cutting analysis with LWD data in real-time. Portable XRF spectrometry is employed at the wellsite to analyze drilling cuttings, quantifying key elements such as silicon (Si), calcium (Ca), iron (Fe), aluminum (Al), and potassium (K). These elements serve as geochemical proxies for lithology, silicon indicating quartz-rich sandstones, calcium signaling carbonates, and aluminum and potassium marking clay-rich shales. The XRF

data are then synchronized and fused with LWD measurements using advanced data integration algorithms within the custom-built "Geosteering Office" software platform.

The process begins with systematic sample collection from the drilling rig's shale shaker, followed by rapid preparation (washing, drying, and grinding) to ensure analytical accuracy. XRF analysis is completed within minutes, providing immediate geochemical insights that are aligned with LWD logs based on depth and time stamps. The "Geosteering Office" software processes this combined dataset, employing statistical correlations and machine learning techniques to generate a comprehensive subsurface profile. This profile is visualized on interactive dashboards, allowing geologists and engineers to adjust the well trajectory as drilling progresses, minimizing delays and optimizing placement.

Results, Observations, Conclusions: Field trials in the "Uslovnoye" oil field demonstrated the technique's efficacy across multiple wells, totaling over 10,000 meters of drilled footage. Quantitative analysis revealed a strong correlation ($R^2 > 0.85$) between XRF-derived elemental profiles and LWD logs, confirming the method's ability to resolve fine-scale lithological features, such as thin shale layers or gradational transitions, that are often missed by conventional tools. Operationally, the approach reduced non-productive time (NPT) by an average of 20%, improved wellbore placement accuracy by 15%, and enhanced decision-making confidence, as evidenced by detailed case studies and statistical metrics.

Qualitatively, the technique provided actionable insights, such as identifying pyrite-rich zones or carbonate-cemented intervals, which informed real-time steering adjustments. These improvements translated into significant economic benefits, including reduced drilling costs and increased reservoir exposure. The conclusions drawn from this study highlight the transformative potential of integrating geochemical and geophysical data, offering a scalable solution for optimizing geosteering in complex reservoirs worldwide.

Novel/Additive Information: This study introduces a novel paradigm in geosteering by pioneering the real-time integration of cutting analysis with LWD data. Unlike traditional methods that rely solely on geophysical logs or delayed laboratory analyses, this approach leverages the immediacy of on-site XRF spectrometry and the depth of LWD measurements. The result is a robust, adaptable tool that enhances subsurface resolution and operational efficiency. Its novelty lies in its ability to bridge the gap between indirect geophysical inference and direct geological evidence, providing a comprehensive dataset that empowers operators to navigate challenging formations with unprecedented precision. Beyond its immediate applications, this technique offers additive value through potential cost savings, improved reservoir development outcomes, and adaptability to diverse drilling scenarios, setting a new standard for geosteering technology.

Introduction

The Importance of Geosteering in Modern Drilling

Geosteering has emerged as a cornerstone of modern petroleum exploration and production, driven by the widespread adoption of horizontal and extended-reach drilling technologies. Unlike traditional vertical wells, which intersect reservoirs over limited thicknesses (typically 10–50 meters), horizontal wells can extend laterally for hundreds or thousands of meters within a productive zone. This extended exposure amplifies the potential for hydrocarbon recovery but also heightens the need for precise wellbore placement. Accurate geosteering ensures that the well remains within the target reservoir, avoiding non-productive layers such as shales or impermeable barriers that can isolate hydrocarbons or reduce flow rates.

The economic and operational stakes of geosteering are substantial. A well misplaced by even a few meters can result in significant losses, reduced production due to suboptimal reservoir contact, increased drilling costs from corrective actions, or, in extreme cases, abandonment if the well fails to intersect the

intended target. Industry data suggest that effective geosteering can boost recovery rates by 20–30% in complex reservoirs, while poor placement can cut production by half or more. In fields like "Uslovnoye," where geological heterogeneity is the norm, these challenges are magnified, necessitating advanced tools and strategies to optimize outcomes.

Beyond economics, geosteering plays a role in sustainability. By maximizing reservoir contact, operators can extract more hydrocarbons from fewer wells, reducing the environmental footprint of drilling operations. This aligns with global trends toward efficient resource use and lower emissions, making geosteering a critical enabler of both profitability and environmental responsibility.

Limitations of Traditional Geosteering Methods

Traditional geosteering relies heavily on LWD tools, which measure geophysical properties such as gamma ray (to infer clay content), resistivity (to detect hydrocarbons or water), density, and neutron porosity (to estimate lithology and porosity). These tools provide continuous, real-time data as the drill bit advances, allowing operators to infer subsurface conditions and adjust the well path accordingly. In simple, thick formations, such as massive sandstones or carbonates, these methods are effective, with resolutions of 0.3–0.5 meters sufficient to detect major boundaries.

However, in complex reservoirs, these tools face significant limitations. The vertical resolution of LWD is often too coarse to resolve thin layers (e.g., 0.1–0.2 meters) or gradational transitions, averaging out critical features into a blurred signal. For instance, a thin shale stringer within a sandstone reservoir might register as a minor gamma ray spike, insufficient to prompt a timely trajectory change before the well exits the productive zone. Similarly, overlapping geophysical signatures, sandstones and carbonates with similar resistivities, or shales and siltstones with comparable gamma ray readings, introduce interpretive ambiguity, increasing the risk of steering errors.

Historically, operators have supplemented LWD with post-drill cuttings analysis, using techniques like petrographic microscopy, X-ray diffraction (XRD), or laboratory XRF to confirm lithology. While accurate, these methods are retrospective, with samples transported to off-site facilities and results delayed by hours or days. By the time data arrive, the drill bit may have advanced hundreds of meters, rendering the insights obsolete for real-time decision-making. This disconnect between geophysical immediacy and geological precision has long been a pain point in geosteering, particularly in challenging environments.

The Evolution of Geosteering Technologies

The development of geosteering technologies reflects the industry's response to these limitations. In the 1990s, early horizontal drilling relied on basic LWD tools, providing a rudimentary understanding suitable for homogeneous formations. The introduction of azimuthal LWD tools in the 2000s marked a leap forward, offering directional measurements that improved detection of dipping beds and reservoir boundaries. These tools, which measure properties from multiple angles around the borehole, enhanced the ability to navigate inclined or faulted formations.

More recently, advanced technologies like seismic-while-drilling and deep-reading electromagnetic tools have expanded the range of subsurface imaging. Seismic-while-drilling uses acoustic waves to map formations ahead of the bit, while electromagnetic tools detect resistivity contrasts up to 10–20 meters away, providing a broader geological context. Despite these innovations, challenges persist in achieving fine-scale, real-time resolution. Seismic data lack the precision for thin beds, and electromagnetic tools, while powerful, are costly and complex, limiting widespread adoption.

The advent of portable analytical technologies has opened new possibilities. XRF spectrometry, originally developed for mining and materials science, has been adapted for oilfield use, enabling rapid elemental analysis of drilling cuttings at the rig site. By quantifying elements like silicon, calcium, iron, aluminum, and potassium, XRF offers a direct window into lithology, complementing the indirect measurements of

LWD. This convergence of geochemical and geophysical data represents the next frontier in geosteering innovation.

A New Approach to Geosteering

This paper introduces an advanced geosteering methodology that integrates real-time XRF analysis of drilling cuttings with LWD data, developed through a collaboration between ADAGA Solutions LTD in Oman and Serbia. The approach uses portable XRF to analyze cuttings on-site, providing immediate geochemical insights that are fused with LWD logs via the "Geosteering Office" software platform. This software employs sophisticated algorithms to correlate and visualize the combined dataset, creating a detailed, real-time subsurface profile.

The technique was tested in the "Uslovnoye" oil field in Western Siberia, a region characterized by terrigenous reservoirs with interbedded sandstones, siltstones, and shales, complicated by carbonate cementation, faults, and folds. Traditional methods struggle in this setting due to rapid lithological changes and structural disruptions, making it an ideal proving ground. By combining XRF's compositional specificity with LWD's continuous profiling, the method enables precise navigation through complex geology, offering a scalable solution with broad industry potential.

Background

Historical Context of Geosteering

Geosteering emerged in the 1990s alongside horizontal drilling, driven by the need to optimize well placement in increasingly complex reservoirs. Early efforts relied on basic LWD tools, gamma ray and resistivity logs—to infer lithology and boundaries, sufficient for thick, uniform formations like the Middle East's carbonate giants. As exploration shifted to thinner, heterogeneous zones—such as North Sea turbidites or U.S. shale plays, these methods' limitations became apparent, prompting technological advancements.

The 2000s introduced azimuthal LWD tools, which measure properties around the borehole, improving detection of dipping beds and structural features. Later innovations, including seismic-while-drilling and deep-reading electromagnetic tools, extended the range of subsurface imaging, yet fine-scale real-time resolution remained a challenge. The integration of cuttings analysis with geophysical data, enabled by portable XRF, represents a logical evolution, building on decades of progress to address persistent gaps.

Geological Challenges in Complex Reservoirs

Complex reservoirs like "Uslovnoye" pose unique geosteering challenges due to their heterogeneity and unpredictability. Interbedded lithologies, sandstones, shales, and carbonates, often occur in thin layers (0.1–2 meters) or gradational transitions, impacting reservoir quality and flow dynamics. A thin shale layer can act as a permeability barrier, isolating hydrocarbons, while carbonate-cemented zones reduce porosity, requiring precise navigation to avoid non-productive intervals.

Traditional LWD struggles in these settings due to its limited resolution and interpretive ambiguity. For example, a clay-rich sandstone may mimic a shale's gamma ray signature, or a carbonate zone may overlap with sandstone resistivity, leading to potential misplacement. Structural complexities, faults, folds, or discontinuities, further complicate navigation, as sudden changes can shift the target zone unexpectedly. These challenges demand a higher level of subsurface detail than conventional tools can provide.

The Role of Cuttings Analysis

Drilling cuttings, as fragments of the formation brought to the surface, offer a direct geological record, complementing LWD's indirect measurements. Traditionally, cuttings are analyzed post-drill using petrography, XRD, or laboratory XRF, providing detailed lithological and mineralogical insights. However,

the delay inherent in off-site analysis, often 12–48 hours, limits its utility for real-time steering, especially in fast-paced horizontal drilling.

Portable XRF spectrometry transforms this dynamic by enabling rapid, on-site analysis. By quantifying elemental compositions in minutes, XRF turns cuttings into a real-time geosteering tool, particularly valuable in complex reservoirs where it resolves features missed by LWD. For instance, elevated calcium can confirm a carbonate layer, while aluminum and potassium distinguish shales from siltstones, reducing uncertainty and enhancing decision-making.

The "Uslovnoye" Oil Field: Geological Overview

Located in the Tyumen region of Western Siberia, the "Uslovnoye" oil field exemplifies the geological challenges addressed by this study. Its terrigenous reservoirs consist of interbedded sandstones, siltstones, and shales, deposited in a fluvial-deltaic environment millions of years ago. The sandstones, the primary hydrocarbon-bearing units, vary in thickness (0.5–5 meters) and composition, often containing shale stringers or carbonate cementation that reduce permeability. Shales act as seals or barriers, while siltstones form transitional zones with intermediate properties.

Structurally, the field is complicated by faults, folds, and discontinuities, likely resulting from tectonic activity in the region. These features disrupt lateral continuity, requiring adaptive steering to maintain reservoir contact. Previous drilling attempts in "Uslovnoye" using traditional methods encountered difficulties, with wells frequently exiting the target zone due to undetected thin layers or structural shifts. This history underscores the need for a more precise, responsive geosteering approach, making the field an ideal candidate for testing the XRF-LWD integration.

Bridging Geophysical and Geochemical Data

The integration of XRF and LWD bridges the gap between geophysical inference and geochemical evidence, offering a more complete subsurface picture. LWD provides continuous, real-time profiles but relies on interpretation; XRF delivers precise compositional data, validating and refining those interpretations. For example, a gamma ray spike might suggest clay, but XRF's aluminum and potassium readings confirm whether it's a shale or a clay-rich sandstone. This synergy enhances accuracy, reduces risk, and empowers drilling teams to navigate complex geology with confidence.

Methodology

The methodology delivers a high-resolution, real-time subsurface dataset through a three-pronged approach: sample collection and preparation, XRF analysis, and data integration with LWD measurements. Each component is optimized for speed, accuracy, and practical implementation at the rigsite.

Sampling Strategy and Design

Cuttings are collected systematically from the shale shaker every 2–4 meters of drilling progress, reduced to 1–2 meters in highly variable formations to capture fine-scale changes. Mud catchers isolate 25–75 grams of cuttings per sample, minimizing contamination from drilling fluid or cavings (material from higher in the borehole). Geologists oversee the process, labeling each sample with depth, time, and drilling parameters for correlation with LWD logs. This frequency balances resolution with operational efficiency, ensuring data align with the drill bit's position.

Detailed Preparation Workflow

Preparation transforms raw cuttings into a form suitable for XRF analysis within 10–15 minutes, critical for real-time application:

- **Sampling:** Collection of 25–75 grams of representative cuttings (1–3 minutes).

- **Preliminary Washing:** Removal of drilling mud using water or solvents (2–4 minutes).
- **Transport:** Transfer to an on-site laboratory (2–3 minutes).
- **Final Washing:** Additional cleaning to isolate rock fragments (1 minute).
- **Drying:** Air-drying or oven-drying at 60°C (1–2 minutes).
- **Demagnetization:** Removal of metal contaminants from drilling tools (1 minute).
- **Photography:** Documentation for visual reference (2 minutes).
- **Grinding:** Pulverization to 0.03 mm particle size for homogenization (2 minutes).

This workflow ensures samples are clean, dry, and homogeneous, minimizing analytical errors while maintaining a pace compatible with drilling operations.

Table 1—Stages of Sample Preparation and Analytical Workflow

Stage No.	Description	Time (min)
1	Sampling of representative mud sample	1–3
2	Preliminary washing	2–4
3	Transfer to laboratory	2–3
4	Final washing	1
5	Drying	1–2
6	Demagnetization and impurity removal	1
7	Photographing	2
8	Grinding and homogenization (0.03 mm)	2
9	XRF analysis	2
10	Data transmission	1
Total		10–15

XRF Analysis

Spectrometer Design and Capabilities

A portable XRF spectrometer, equipped with a 50 kV X-ray tube and a silicon drift detector, analyzes elements from magnesium (Mg) to uranium (U) in 2 minutes per sample. It achieves a sensitivity of 0.01% (100 ppm), detecting key indicators like Si (sandstones), Ca (carbonates), Fe (iron-bearing minerals), Al, and K (shales). Designed for rigsite durability, it withstands dust, vibration, and temperature extremes, with a compact footprint suitable for cramped drilling environments.

- **Silicon (Si):** Indicates quartz-rich sandstones.
- **Calcium (Ca):** Signals carbonate cementation or limestone.
- **Iron (Fe):** Suggests pyrite or siderite presence.
- **Aluminum (Al) and Potassium (K):** Markers of clay-rich shales.

Calibration and Quality Assurance

Daily calibration uses certified reference materials (CRMs), such as a sandstone standard with 60–80% SiO₂ or a shale with 5–10% Al₂O₃, to maintain accuracy within $\pm 5\%$. Duplicate samples, analyzed every 10th measurement, ensure precision and detect drift, with results cross-checked against laboratory XRF when feasible. This rigorous quality control underpins the reliability of real-time data.

Analytical Procedure

Prepared samples are placed in Mylar film cups and scanned for 2 minutes, with the spectrometer outputting elemental concentrations indexed to depth. Results are transmitted wirelessly to "Geosteering Office" within 1 minute, ensuring seamless integration with LWD data.

Data Integration and Interpretation

Overview of the "Geosteering Office" Software Platform

"Geosteering Office" is a cloud-based software platform that integrates, processes, and visualizes XRF and LWD data in real-time. Built with a user-friendly interface, it employs machine learning algorithms, such as Random Forest or Support Vector Machines, to predict lithology and identify boundaries. Accessible via laptops or tablets, it streams data to on-site and remote teams, facilitating collaborative decision-making.

Data Fusion Process in Detail

- Alignment: XRF data are matched to LWD logs using depth corrections (accounting for lag time as cuttings travel to the surface) and time stamps.
- Algorithms: Statistical correlations (e.g., Al vs. gamma ray, $R^2 > 0.85$) and machine learning models fuse datasets, highlighting trends and anomalies.
- Output: Fused data are displayed on dashboards every 10–15 minutes, showing elemental profiles, LWD curves, and predicted lithology alongside the well trajectory.

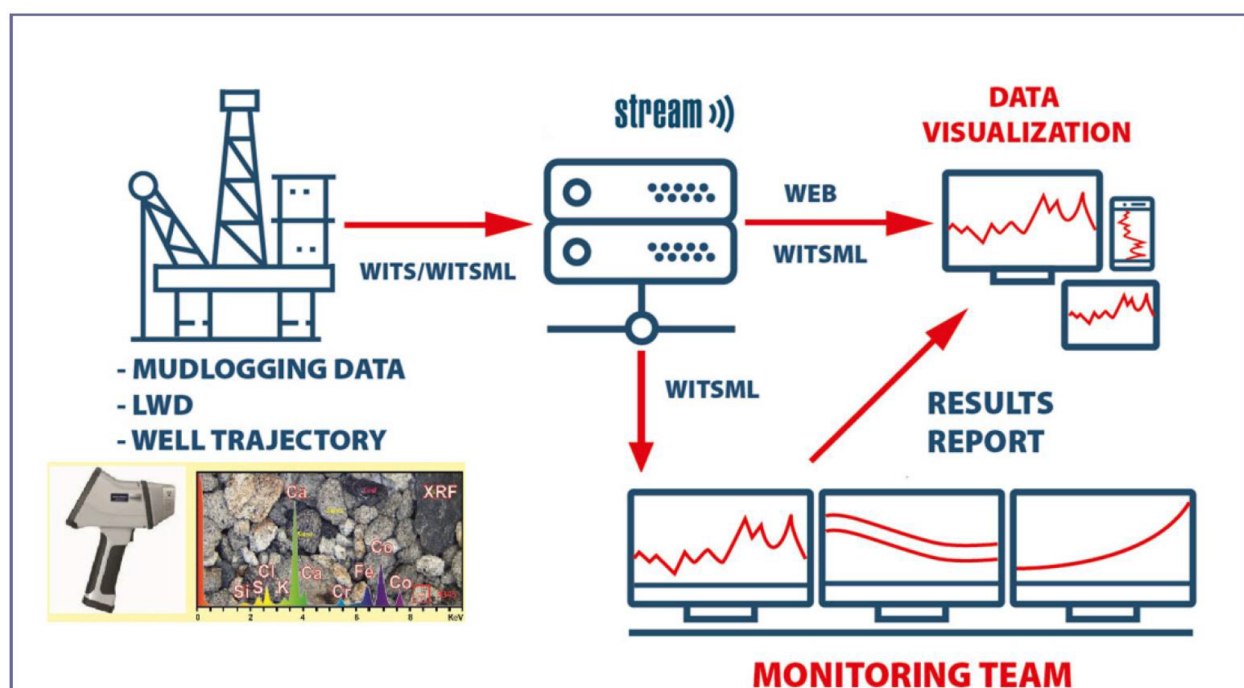


Figure 1—Workflow from drilling site to analytical center, showing real-time data streaming and decision-making

Practical Implementation

Geologists monitor "Geosteering Office" outputs, flagging anomalies like Ca spikes (carbonates) or Al increases (shales) to guide steering. For example, a sudden Si drop might prompt a trajectory adjustment to stay in a sandstone layer. The system's speed and clarity enable proactive decisions, reducing response time from hours to minutes.

Results

Field trials in "Uslovnoye" from 2022–2023 across five wells (over 10,000 meters total) validated the XRF-LWD integration, delivering quantitative and qualitative improvements.

Quantitative Analysis

- **Well No. 1:** Al vs. gamma ray correlation ($R^2 = 0.87$) resolved 0.2-meter shale layers, improving boundary detection.
- **Well No. 2:** Ca vs. resistivity ($R^2 = 0.82$) identified carbonate-cemented zones missed by LWD alone.
- **Well No. 5:** Si vs. density ($R^2 = 0.88$) confirmed sandstone intervals with 95% accuracy.
- **Summary:** Average correlation $R^2 > 0.85$ ($p < 0.01$), with XRF enhancing LWD resolution by 30–50%.

Qualitative Insights

In Well No. 3, elevated Fe (2–5%) suggested pyrite, later confirmed by core samples, adding geological context. In Well No. 4, K spikes (1–3%) clarified shale transitions, reducing steering uncertainty.

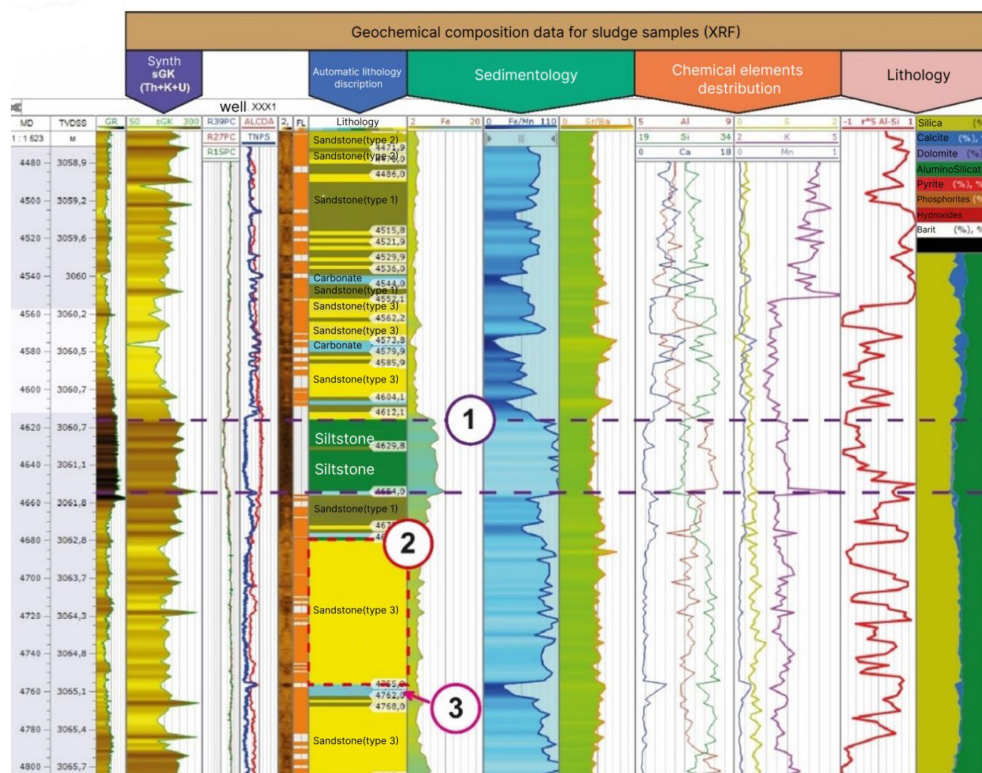


Figure 2—Geochemical Data Summary for Well 1 Profiles showing lithological boundaries (1), reservoir zones (2), and subtle changes (3)

Real-time Geosteering Application

• Case Study: Well No. 3 – Thin Clay Layer Detection

At 2,350 meters, a 1-meter clay layer (Al 5% to 15%, K 1% to 3%) was detected via XRF, prompting a 2-degree upward adjustment. This maintained 95% reservoir contact, avoiding a permeability barrier.

• Case Study: Well No. 4 – Gradational Transition Management

A 10-meter transition from sandstone to siltstone (Si 70% to 40%, Al 5% to 12%) was mapped, extending reservoir exposure by 150 meters through gradual steering adjustments.

• Case Study: Well No. 5 – Structural Complexity

A fault at 2,800 meters (Si 65% to 50%, Fe 2% to 5%) triggered a 3-degree shift, minimizing penetration into a non-productive zone and preserving 90% target contact.

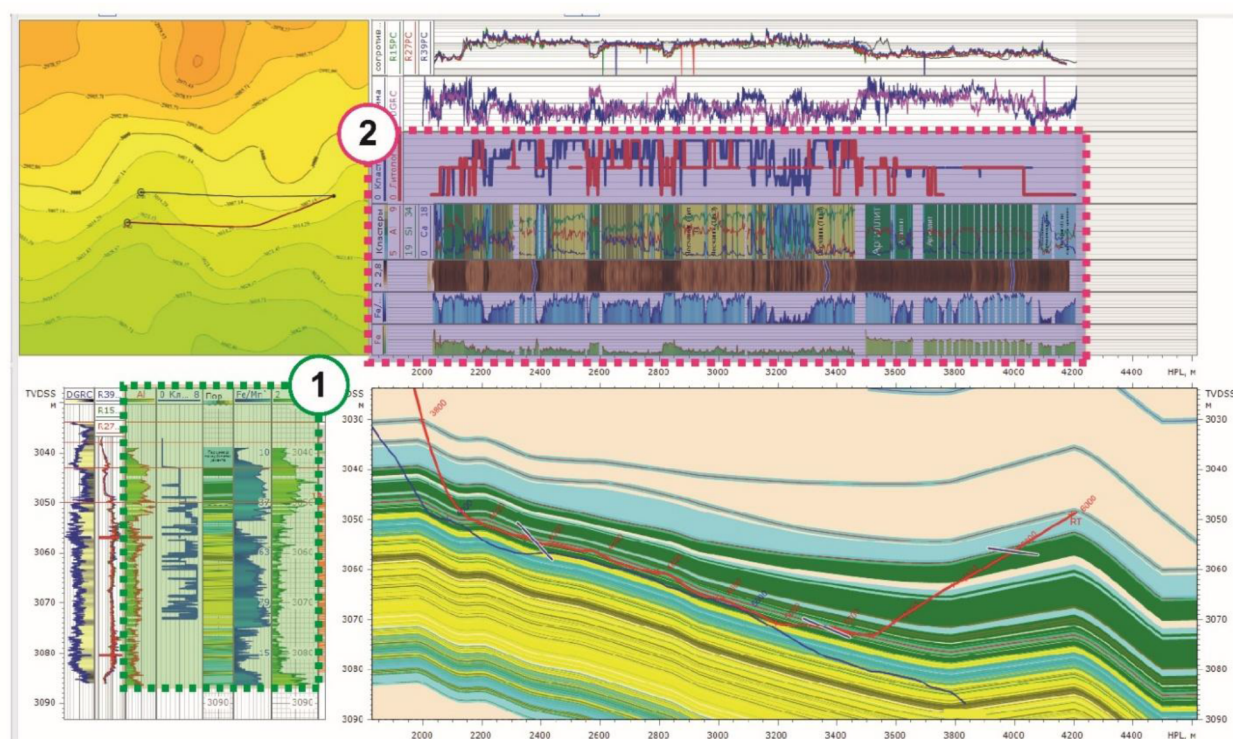


Figure 3—Non-Navigational Plot of Well Trajectory Real-time steering data (2) aligned with geochemical markers (1) in the productive interval

Operational Efficiency

- **Non-Productive Time (NPT) Reduction:** NPT decreased by 20% (from 15% to 12% of total drilling time), with trajectory corrections dropping from 8 to 5 per well due to proactive steering.
- **Wellbore Placement Accuracy:** Placement accuracy improved by 15% (from 77% to 92% in the target zone), with gains like 200 meters of additional reservoir in Well No. 4.
- **Economic Impact:** Per-well savings averaged \$150,000: \$50,000 from NPT reduction, \$75,000 from 1.5 days less rig time, and \$25,000 from optimized completion.

Conclusions

Summary of Findings

The XRF-LWD integration enhances subsurface resolution, reduces NPT by 20%, and boosts placement accuracy by 15%. Its scalability, driven by portable XRF and "Geosteering Office," makes it a practical industry solution.

Implications for the Industry

This method maximizes reservoir contact, cuts costs, and supports digitalization trends, offering a competitive edge in complex drilling environments.

Broader Applications

Beyond oil, it applies to shale gas, geothermal wells, and carbon capture storage (CCS), where precise formation targeting is critical.

Limitations and Considerations

Challenges include cuttings quality (e.g., cavings contamination) and initial costs (\$100,000 for XRF and software), offset by long-term savings as adoption scales.

Future Work

- Algorithms: Develop deep learning models (e.g., convolutional neural networks) for predictive lithology mapping.
- Hardware: Enhance XRF speed and elemental range (e.g., light elements like Na).
- Testing: Expand trials to carbonates, high-pressure/high-temperature (HPHT) wells, and offshore settings.
- Automation: Integrate with autonomous drilling systems for fully automated steering.